

Social Network Influences on Adolescent Substance Use

Sunbelt
INSNA 2026
Daytona Beach, FL, USA

E-cigarette Cessation as a Disadoption Process: Social Network Influences on Stable and Temporary Quitting

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Data Collection

Wave	1	2	3	4	5	6	7	8
Time	Fall 2020	Spring 2021	Fall 2021	Spring 2022	Fall 2022	Spring 2023	Fall 2023	Spring 2024
Grade-C1	9th	9th	10th	10th	11th	11th	12th	12th
Grade-C2			9th	9th	10th	10th	11th	11th
N Schools	5	11	14	14	14	14	14	14
N Students	852	2,240	3,525	3,539	3,379	3,123	2,097	1919

No. of schools/networks	100	11 Unique Schools
No. of students	23,012	

Network Question (Roster Pre-loaded)

Name up to seven (7) of your closest friends in your grade at school in the spaces below. Enter your friend's first and last real name, not their nickname. For example, if you call your friend "Tony" but his real name is "Anthony Smith," enter "Anthony Smith." As you enter the first few letters of their first name, possible choices will automatically appear on the screen. Please select the friend you want to name. If no name matches, just enter the name you wish to enter.

Outcomes:

- Past 6-month use of E-cigarettes
- Also assessed Ever-use
- Also asked about other substances ...

Covariate Controls

- Grade in school
- Cohort (2025 vs. 2024)
- Female identity
- Sexual minority
- Financial precariousness
- Latine identity
- Asian

Results: Demographics

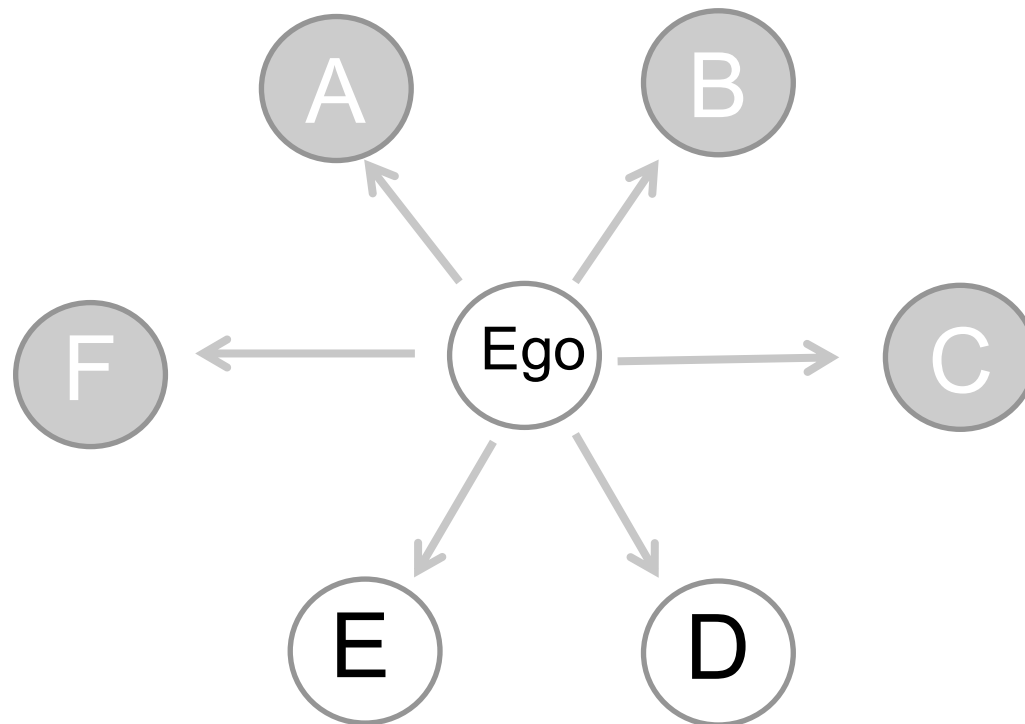
% Female	52.3
% Financial Precariousness	39.4
% Sexual Minority	6.8
Parental Education	4.85
% Latine	44.3
% Other	33.4
% Asian	34.5
% Multi	17.9
% White	14.2

Network Data

Perceived Use Somewhat Higher than Friend Self-report

	Mean	Standard Deviation	Range
Out Degree (# friends named)	2.58	1.97	0 - 7
In Degree (# times named as friend)	2.58	2.21	0 - 16
# Friends Self-report E-cigarette Use	0.34	0.69	0 - 5
# Friends Self-report Marijuana Use	0.23	0.58	0 - 6
# Friends Self-report Alcohol Use	0.58	0.93	0 - 6
# Friends Self-report Any	0.38	1.95	0 - 17
# Friends Perceived Use E-cigarettes	0.48	1.07	0 - 5
# Friends Perceived Use Marijuana	0.47	1.11	0 - 5
# Friends Perceived Use Alcohol	0.72	1.35	0 - 5
# Friends Perceived Use Any	0.57	1.04	0 - 5

Exposure is Associated with Adoption



Networks

	E-cigarette	Marijuana	Alcohol	Any
Out-degree	0.73*	0.79*	0.81*	0.87*
In-degree	0.97	0.99	0.93	0.97
# Friends Self-report Use	2.08*	2.38*	1.83*	1.44*
# Friends Perceived Use	1.67*	1.94*	1.51*	2.14*
# Asian Friends Named	0.69	0.53	0.70	0.68
# Named as Friend by Asians	0.38*	0.29*	0.55*	0.42*

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Social network influences on adolescent substance use: Exposure, perceptions, and ethnicity

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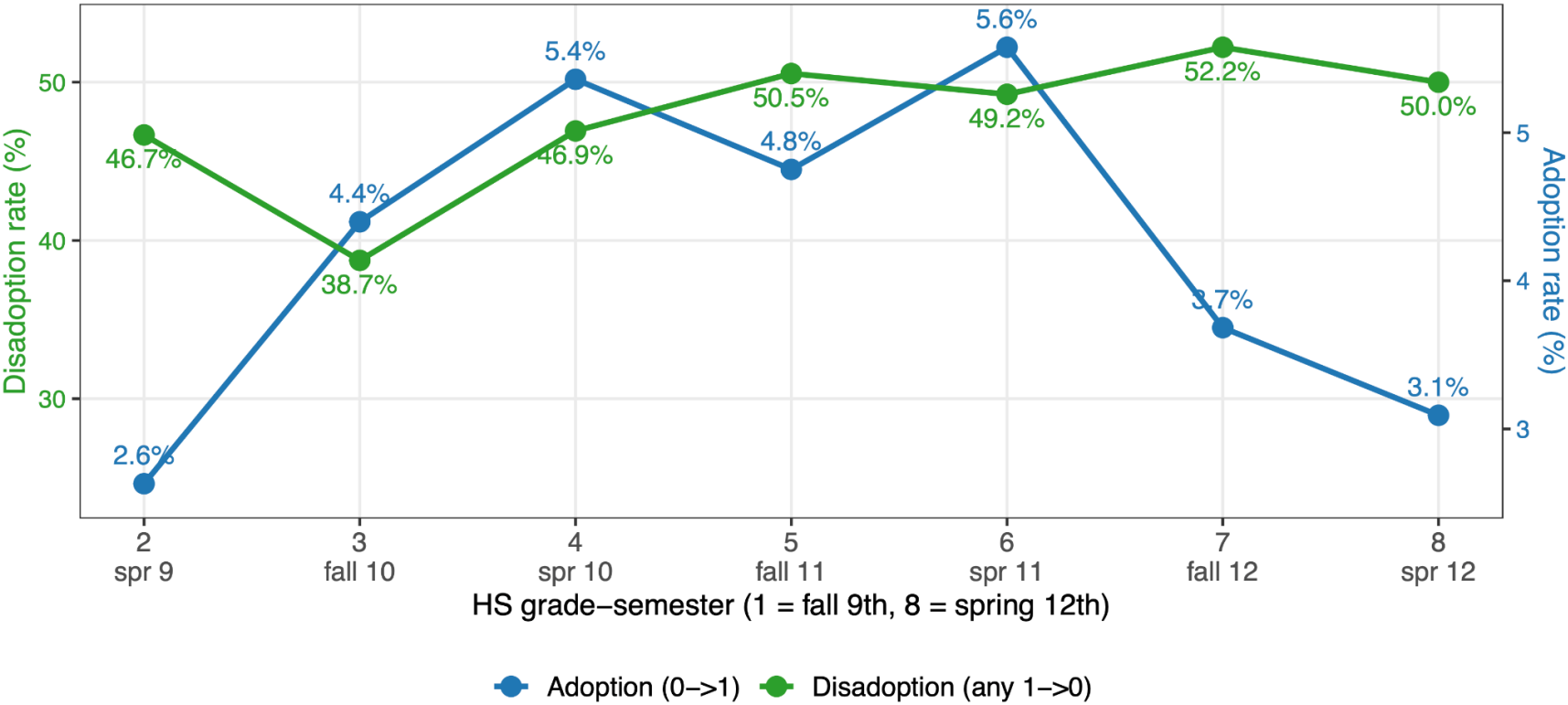
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Quitting

- Many studies have shown that individual substance use is associated with peer use.
- Much less frequently studied is quitting or dis-adoption
 - Stable quitting, Use $\longrightarrow$ Quit
 - Temporary quitting, Use $\longrightarrow$ Quit ...
 - Unstable quitting, Use $\longrightarrow$ Quit $\longrightarrow$ Use

Adoption and disadoption trajectories across the high-school years



Past 6-Month E-cigarette Use

	Adopters	Stable Quitters	Temporary Quitters	Unstable Quitters
N Students	1,711	161	182	161
N Events	193	89	129	15
Sexual Minority	1.45*	0.68	0.84	1.24
Asian	0.64*	0.91	0.98	0.35
Major Depressive Symptoms	1.17	0.74	0.44*	0.21*
Generalized Anxiety Disorder	0.87	1.43	1.20	1.36
Out-degree	0.91	1.18	0.94	0.91
In-degree	1.10*	1.09	1.13	1.07
Perceived Friend Use (N)	1.47*	0.73*	0.72*	0.78
Network Exposure Users	5.68*	0.10*	0.23*	0.80
Network Exposure Disadopters	1.81	0.37	0.40	0.04

Quit rates as a function of the number of perceived friends who vape

# Perc. Friends	At Risk for Use	N Users	Use Rate	N at Risk for Quitting	N Quitters	Quit Rate
0	17492	376	0.0215	388	210	54.12%
1	2067	106	0.0513	224	95	42.41%
2	1246	111	0.0891	247	77	31.17%
3	652	62	0.0951	223	61	27.35%
4	214	20	0.0935	127	27	21.26%
5	269	31	0.1152	193	40	20.73%

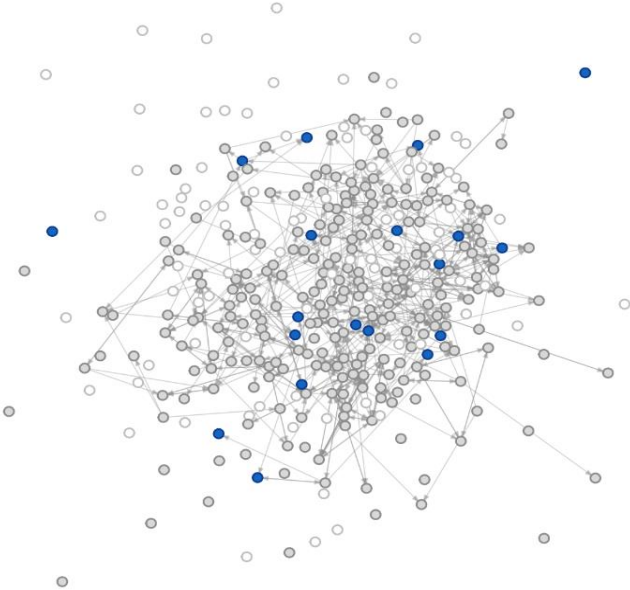
Quit rates as a function of the number of network ties who vape

# Network Friends	At Risk for Use	N Users	Use Rate	N at Risk for Quitting	N Quitters	Quit Rate
0	12223	384	0.0314	537	280	52.14%
1	2024	154	0.0761	285	130	45.61%
2	342	31	0.0906	91	34	37.36%
3	45	6	0.1333	17	5	29.41%
4	6	1	0.1667	5	2	40.00%
5+	1	0	0	2	0	0.00%

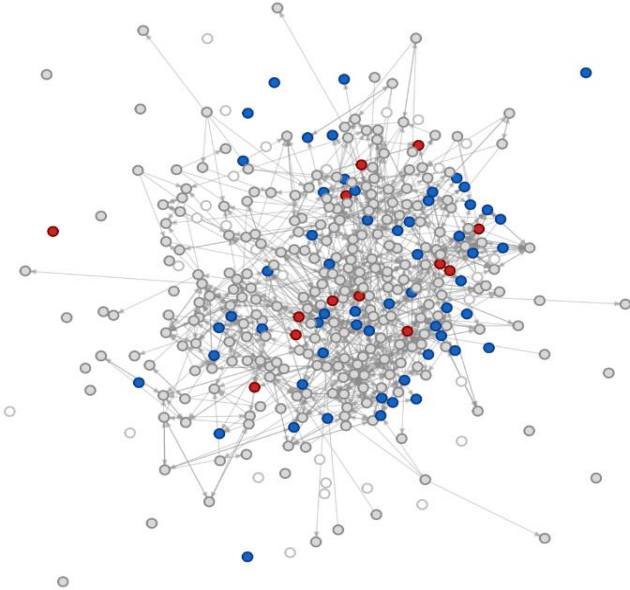
Conclusions

- Use was associated with peer use as usual
- No Demographic variables were associated with quitting either stable or temporary
- Major depressive symptoms was associated with temporary quitting

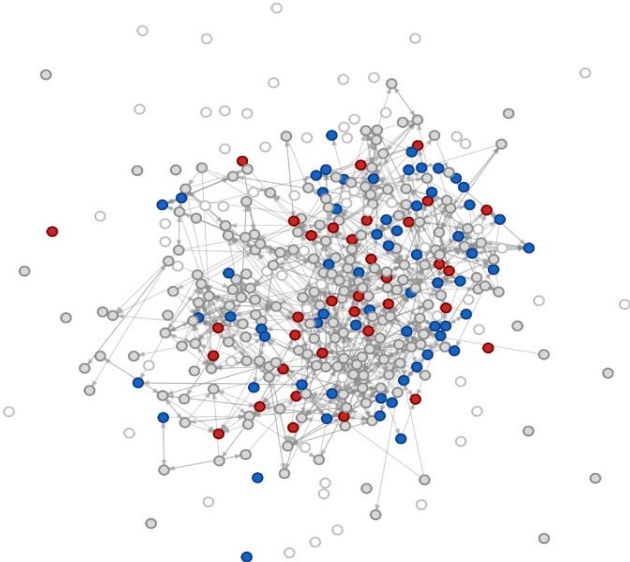
Spring 9th grade



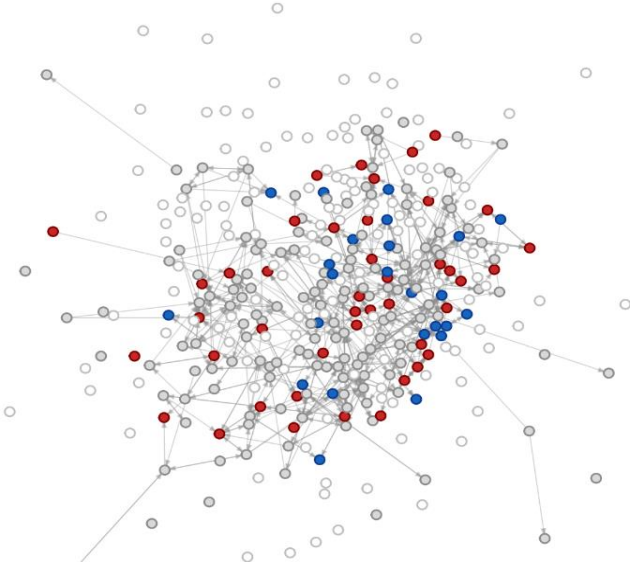
Spring 10th grade



Spring 11th grade



Spring 12th grade



- School 114
- Current user (ecig=1)
 - Disadopter (was 1, now 0)
 - Never-user
 - Missing (NA)

Conclusions (2)

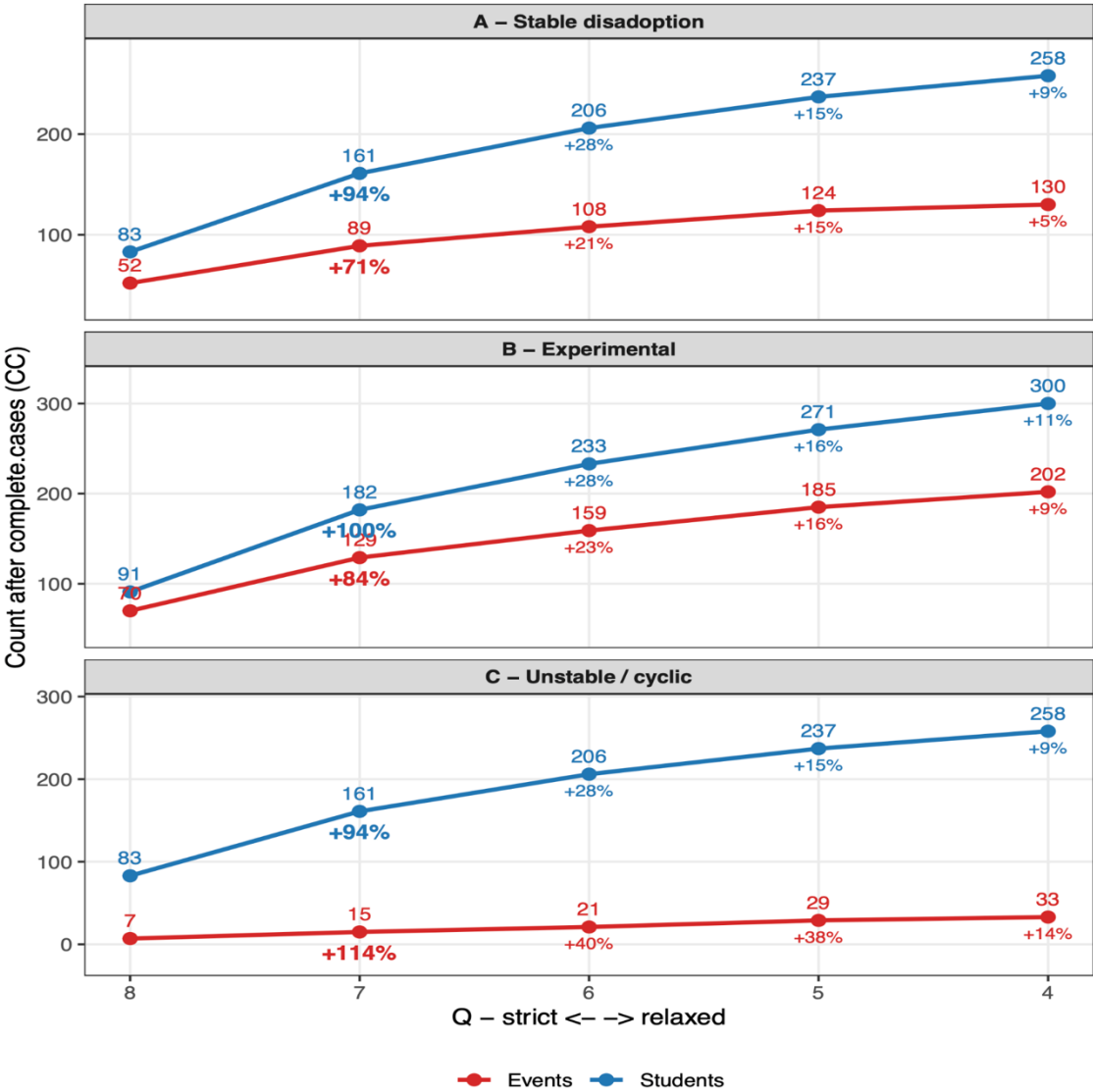
- Both perceived network friend use were inversely associated with quitting whether stable or temporary
- Exposure to quitters (disadopters) was not associated with quitting

Discussion

- This suggests that teens will continue using e-cigarettes when their friends use them but will discontinue if they do not have continued support to use

Thank You!

Q sensitivity: how N students and N events shrink as Q tightens



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Asian Students More Susceptible to Peer Influence

Table 4. Adjusted odds ratios for peer influences on substance use separately by Asian and Non-Asian Students.†

	E-cigarette	Marijuana	Alcohol	Any
<u>Asian Students</u>				
# Friends Self-report Use	2.83**	4.67**	2.23**	5.34**
# Friends Perceived to Use	1.96**	3.03**	2.03**	2.91**
<i>Non -Asian Students</i>				
# Friends Self-report Use	1.84**	2.20**	1.84**	2.90**
# Friends Perceived to Use	1.63**	1.88**	1.45**	2.16**
<u>Wald Test Z Scores</u>				
# Friends Self-report Use	1.70*	2.46**	1.12	2.03*
# Friends Perceived to Use	1.19	2.61**	2.96**	1.74*

Additionally

- Results replicated on past 6 months use
- Results did not differ by whether the school was Asian majority.

Other Publications

- Piombo, S. E., Vega Yon, G., & **Valente, T. W.** (2025). The impact of social norms on diffusion dynamics: A simulation of e-cigarette use behavior. Health Education and Behavior, 00, doi:10.1177/10901981251327189
- Piombo, S. E., de la Haye, K., & **Valente, T. W.** (2025). Network dynamics of social influence on e-cigarette use among an ethnically diverse adolescent cohort. Nicotine and Tobacco Research, 00,
- Piombo, S. E., Barrington-Trimis, J., & **Valente, T. W.** (2023). Network norms and individual electronic nicotine delivery systems use. BMC Public Health.